[METRIC]

A-A-2549C
July 11, 1996
SUPERSEDING
A-A-2549B
February 1, 1996

COMMERCIAL ITEM DESCRIPTION

BINDER, LOOSE-LEAF (RING)

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description covers loose-leaf ring binders. These binders are used to retain, organize and protect paperwork.

2. CLASSIFICATION. The binders shall be of the following grades as specified:

Grade I - Cloth bound, Flexible cover
Grade II - Cloth bound, Stiff cover
Grade III - Plastic bound, Flexible cover
Grade IV - Plastic bound, Stiff cover

Sheet dimension, color, capacity, number of rings and ring spacing center to center shall be as specified in the contract or purchase order.

3 SALIENT CHARACTERISTICS:

3.1. Metal. The metal portion of the binder, as used herein, refers to the steel parts of the binder, which include the metal hinges, backplate, cover plate, rings, ring plate, lever and rivets. The exposed metal portion of the binder includes all parts cited above which are visible when an empty fully assembled binder is examined. All metal components, except rings, shall be of the thickness and hardness specified in Table III.

3.2. Rings (Arches). Rings shall be made of steel with the diameter and hardness in accordance with Table IV.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document, should be sent to: General Services Administration, Federal Supply Service, 26 Federal Plaza, New York N.Y. 10278 ATTN: Engineering Branch (2FYEE).

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3.3. Metal Finish. The exposed metal parts and components shall be corrosion and moisture resistant, and coated in accordance with normal commercial practice providing a bright metal finish.

3.4. Coating Compound. The coating compound for all Grade I and II binders shall be any plastic, commercially acceptable for the purpose intended. Plasticizer shall be adequate to provide flexibility and resistance to drying and cracking of the finished material. Unless otherwise specified, the coated woven and coated non-woven covers shall have a levant grain finish.

3.5. Cover Material.

3.5.1. Grade I & II Binders. Cover material for Grade I & II binders shall be any of the following materials commercially acceptable for the purpose intended that meet the requirements specified herein: coated woven cotton, coated woven cotton polyester, or coated non-wovens.

3.5.2. Coated Woven Cloths. Coated woven cotton cloth, and coated woven cotton polyester cloth shall consist of the base cloth saturated with the coating compound. The base cloth shall be evenly coated with the coating compound and shall be free from defects, such as fish eyes, holes, and cracks. The coated woven cloth shall have a colorfastness of at least "good", when tested for crocking in accordance with Fed.Std.191/Method 5651. The cover material for Grade I & II binders shall meet the requirements of Table V.

3.5.3. Coated Non-Woven Covers. For the purpose of this specification, the definition of non-woven by the National Association of State Textbook Administration (NASTA) shall be accepted. Either of the following non-woven materials shall be acceptable.

(NASTA) Type II - Reinforced Paper: Polymer or Resin-reinforced Cellulose Fiber Webs.
(NASTA) Type III - Synthetic-fiber Structure: Strands of Interconnected Fiber.

NOTE: All NASTA material shall be in conformance with the NASTA specification as cited in the "Manufacturing Standards and Specification for Textbooks".

For Nasta Type II & III non-woven, thickness of the substrate (uncoated NASTA Type II or III material) shall be sufficient to meet the requirements of Specification NASTA Type II or III. The plastic coating shall be in accordance with normal commercial practice and shall have a minimum thickness of 0.0127 mm (0.0005").

3.6 Cover Board. The cover board shall conform to the requirements of Table VI.

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3.7. Design and Construction. All binders shall be designed to provide a secure place for loose-leaf (filler) sheets, and permit one or more sheets to be inserted or removed without having to disturb the remaining contents. The binders shall open on the side or on the end, as specified. The binders shall be of such design to assure that the metal parts or components will not mar the surface of the desks, tables, or shelves. All binders shall have neatly rounded and uniform corners and shall be of the suitable size to protect the size paper specified. All binders shall have the operating mechanism firmly secured through the entire thickness of the back plate with at least two rivets, which shall not be loose, and the covers shall be connected to the back plate by means of the covering material and, when applicable, with the hinge reinforcing strips, or lining material. The back plate shall be constructed of metal or fiberboard, at the supplier's option.

3.7.1. Metal Back Plate. The backplate must meet the requirements of Table III. If metal is used for Grade I & II binders of 2.5 cm (1") and greater capacity, the metal shall be covered on the entire outside surface (surface facing cover material) by not less than 0.64 mm (0.025") thick paperboard or other commercially acceptable material. The inner surface of the back plate may also be covered with paperboard. At the supplier's option, the inside (surface facing rings) may be covered with paperboard not less than 0.64 mm (0.025") thick, or other commercially acceptable material.

3.7.2. Paperboard Back Plate. The back plate shall extend the whole length of the spine within a tolerance of ± 1.6 mm (± 1/16") from the top and bottom. For grades III & IV, if the rivets are concealed, the entire outside surface of the backplate (surface facing cover material) shall be covered with a strip of paperboard in accordance with commercial practice and of sufficient thickness to prevent damage to the cover material by direct contact with rivets. The paperboard back plate shall conform to Table I below.

Table I.

Binder Capacity	Paperboard Thickness	Basis Weight
3.8 cm (1.5") or less	2.54, +0.25, -0.13 mm (0.100, +0.010, -0.005)"	1.67 kg/m^2 ± 5% (342 lb$_m$/1000 ft^2, ± 5%)
greater than 3.8 cm (1.5")	3.18, +0.38, -0.13 mm (0.120, +0.015, -0.005)"	2.10 kg/m^2 ± 5% (422 lb$_m$/1000 ft^2, ± 5%)
Test Methods	TAPPI T 411	TAPPI T 410

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3.7.3. Operating Mechanism. The metal requirements of the operating mechanism shall be as specified in Tables III & IV. The rings shall be securely and permanently attached to the ring plates, or be an integral part of the ring plates. When in the closed position, the rings shall align and fit snugly and securely. The operating mechanism shall have a booster at both ends; (1) to open the rings or, (2) to open and close the rings or, (3) to open, close and lock the rings, (as specified in the contract or purchase order). When in the locked position, a binder equipped with a locking device shall not be capable of being open manually without bending the ring or damaging the mechanism.

3.7.4. Grade I - Coated Cloth Bound Flexible Covers Binders. A concealed paperboard ring protection strip, not less than 19.0 mm (0.75") wide, shall be fastened to the cover board at the point where the ring comes in contact with the cover, when binder is in closed position (Not applicable on binders of 0.64 cm (1/4") capacity or less). The thickness of the paperboard shall be not less than 0.64 mm (0.025"), when tested in accordance with TAPPI T 411. The outer surface of the binders shall be bound with one piece of coated cloth conforming to the appropriate composition specified in Table V. A coated non-woven may be substituted for both, composition 1 & 2 woven cloths. The covering material shall be securely and neatly glued to the outer surface of the covers and backing, and shall be carried over to the inside surface of the binders for at least 13 mm (0.50"). The covering material shall be securely and neatly pleated or turned, to form uniform rounded corners. One continuous piece of coated cloth, conforming to composition 2, shall be neatly and securely glued to the inside of the exposed cover board, the overlapped coated cloth, and the backing. At the option of the manufacturer, the inside surface of the binder having capacity of 13 mm (0.5") or less, may be lined with one continuous piece of a non-woven substrate. The non-woven substrate shall have a minimum weight of 135 g/m^2 (4 oz/yd^2), when tested in accordance with TAPPI T 410, and a minimum bursting strength of 89 Newtons (20 lb_f) in both directions, when tested in accordance with TAPPI T 807. The lining shall be neatly glued to the inside of the exposed cover boards, the overlapped coated cloth, and glued to the covers.

3.7.5. Grade II - Coated Cloth Bound Stiff Covers. The outer surfaces of the binder shall be bound with one piece of coated cloth conforming to Composition 1. A coated non-woven may be substituted. The covering material shall be securely and neatly glued to the outer surface of the binder for at least 13 mm (0.50"). The covering material shall be securely and neatly pleated or turned to form uniform rounded corners. The binder shall have cloth hinge reinforcing strips (Not applicable to metal hinge binders) which shall be one continuous piece and shall cover the back plate and extend at least 13 mm (0.50") onto the covers. The hinge shall be neatly and securely glued to the specific area of the binder.

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The hinge reinforcement shall be made from a starch filled book cloth or a similar cloth material suitable for the purpose intended. A plastic scrim reinforcement may be used as an alternate to the cloth hinge reinforcement. A paper lining, having a basis weight of 88.8 g/m^2 (60 lb$_m$ based on 25" x 38" - 500 sheets), when tested in accordance with TAPPI T 410, shall be neatly and securely glued to the inside surfaces of the exposed cover boards, the overlapped coated cloth, and glued to the covers. The color of the lining material shall harmonize with the color of the cover material. The lining shall prevent the material which it covers from being visible (i.e. opaque).

3.7.6. Grade III - Plastic Bound Flexible Cover Binders and Grade IV - Plastic Bound Stiff Cover Binder. The outer and inner surface of each of these binders shall be neatly covered with plastic covering material. At each folding point a minimum of a single bar score shall be heat sealed to cover as the hinge. The plastic shall overlap the cover board and the backing, be heat sealed around the edges, and cover all surfaces of the cover board. The plastic cover shall meet the requirements specified in Table II below:

Table II.

Characteristic	Requirement	Test Method
Thickness	0.30-0.51 mm (0.012-0.020)"	Any commercial
Specific Gravity	Black: 1.3 max White pastels: 1.36 or less All other: 1.30-1.36	ASTM D 792 Method A
Seam Tensile Strength	50 % min. of plastic cover material strength	ASTM D 638
Cold Crack @ -17.8±0.6°C (0±1°F)	No evidence of cracking	Fed.Std.191 Method 5874
Dimensional Stability 30 min @ 100°C (212°F) 30 min @ 23°C (73.4°F)	 4% maximum change 1% maximum change	ASTM D 1204
Volatility, 24 Hrs @ 70°C (158°F)	1.5% maximum loss	ASTM D 1203
Tear Resistance	52 N/mm (0.3 lb$_f$/mil)	ASTM D 1004
Taber Stiffness	1.5 gm cm minimum	See note (1)

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Note (1): The Taber Stiffness Test shall be performed on a Taber V-5 Stiffness Tester, using 3.8 x 3.8 cm (1-1/2 x 1-1/2)" test specimens, utilizing a 0-10 test range and 1 cm (0.4)" test length with a 10 unit compensator weight on the pendulum. Tests to be conducted at 23°C (73.4°F).

3.7.7. Outside Plastic Covers for Grade IV Binders. When specified, the outside of each binder shall be additionally covered with a clear plastic cover, heat sealed to the outer edge and the bottom of each cover (front and back), and the bottom spine. The outer cover shall be additionally heat sealed along the length of the hinge of each cover. The top of each cover (front and back) and the top of the spine shall be open to allow a cover sheet to be placed inside the full depth of both covers and a narrow sheet down the full depth of the spine. The clear plastic cover shall be 0.20 mm, +0.04 mm, -0.05 mm, (0.008, -0.0015,+0.002)" thick, and shall be of good commercial quality suitable for the purpose intended. At the supplier's option, two small holes in the plastic cover on the spine shall be allowed where the rivets are drilled into the spine.

3.7.8. Fly Leaves (sheet lifters). Two fly leaves shall be attached in the rings of each binder (greater than 2.5 cm (1") capacity), and shall be made from pressboard, fiberboard, plastic or vulcanized fiber. Three separate and distinct holes shall be a part of each fly leaf so that the fly leaf can be easily fitted onto the rings of the binder. Slits from the edge to the hole of the fly leaf to facilitate attachment of fly leaves to the ring shall not be permitted. The length of the fly leaves shall be approximately the length of the sheet length dimension. The size of the fly leaves shall be sufficient to prevent the sheets from resting directly on the junction of the rings and the ring plate.

3.7.9. Pocket. When specified, the binder shall have a pocket attached to the inside of the front cover. The pocket of Grade I binder shall be made of pliable fiberboard and cover material, and shall be securely and neatly fastened to the cover by the overlap of the outside binding material. The pocket of Grades III and IV shall be not less than one third the width of the sheet size dimension for which the binder is designed.

3.7.10. Metal Hinge. When specified, the binder shall be provided with a continuous piano type metal hinge. The metal shall conform to requirements listed in Table III. Each hinge shall be securely attached to or be a part of the cover in accordance with normal commercial practice.

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3.7.11. Label Holder and Label. When specified the binder shall be provided with a label holder and label. For Grades I & II, the label holder shall be recessed and built into the backbone, or securely attached to the backbone. For Grades III & IV, the label holder shall be at least 70% the width of the spine and not less than 6.4 cm (2-1/2") deep and shall be heat sealed on three sides with the top side open. The upper unsealed side of the pocket shall be located no more than 3.8 cm (1-1/2") from the top of the binder. The label holder shall be provided in a manner such as to not detract from the overall appearance of the binders.

3.8. Marking or lettering of binders. The marking or lettering when required, and the method, location, design, and color shall be as specified.

3.9. Identification Marking. The binders shall be legibly and permanently marked with the name or trademark of the manufacturer on the inside of one cover. When specified, the words "U.S. Government", or "Property of U.S. Government" shall be included in the marking. These markings shall be placed on the inside of either, or both, covers so as to be readily visible upon opening of the binder.

3.10. Workmanship. All hardware and visible components shall be neatly and uniformly located in the binders in accordance with good commercial practice. The binders and all components shall be free from defects that adversely affect appearance and serviceability or present a hazard to the user. No substance shall come off during normal handling or storage.

4. REGULATORY REQUIREMENTS:

4.1 Certification (paperboard). The contractor shall certify that the paperboard used in the manufacture of this item contains a minimum of 80 % Post Consumer Recovered Material (PCRM), in accordance with Code 40 of the Federal Regulations Part 247, Comprehensive Guideline for Procurement of Products Containing Recovered Materials; Final Rule. The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Certification (plastic). The contractor shall certify that the plastic used in the manufacture of this item contains a minimum of 25 % Recovered Material (RM), in accordance with Code 40 of the Federal Regulations Part 247, Comprehensive Guideline for Procurement of Products Containing Recovered Materials; Final Rule. The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter as may be otherwise provided for under the provisions of the contract.

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TABLE III.
BACK PLATES, METAL HINGES AND OPERATING MECHANISM, EXCEPT RINGS.

PARTS AND COMPONENTS	BINDER CAPACITY	THICKNESS OF METAL minimum	METAL HARDNESS minimum	TEST METHOD
Back Plate	Under 2.5 cm (1") 2.5 cm (1") Over 2.5 cm (1")	0.38 mm (0.015") 0.46 mm (0.018") 0.51 mm (0.020")	R30T47.5 " "	ASTM E18 " "
Operating mechanism cover plate	Under 2.5 cm (1") 2.5 cm (1") Over 2.5 cm (1")	0.25 mm (0.010") 0.30 mm (0.012") 0.46 mm (0.018")	R30T60.0 " "	ASTM E18 " "
Ring Plate (Arch Frame)	Under 2.5 cm (1") 2.5 cm (1") Over 2.5 cm (1")	0.64 mm (0.025") 0.76 mm (0.030") 0.89 mm (0.035")	R30T52.0 " "	ASTM E18 " "
Levers which open, or open and close	Under 2.5 cm (1") 2.5 cm (1") Over 2.5 cm (1")	0.76 mm (0.030") 0.76 mm (0.030") 0.76 mm (0.030")	R30T52.0 " "	ASTM E18 " "
Levers which open, close and lock	Under 2.5 cm (1") 2.5 cm (1") or more	1.27 mm (0.050") 1.27 mm (0.050")	R30N58.0 "	ASTM E18 "
Metal Hinge	All	0.51 mm (0.020")	R30T47.5	ASTM E18

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TABLE IV. RINGS

RING CAPACITY	ELLIPTICAL (OVAL) RING Cross section width x length, minimum	ROUND RING DIAMETER minimum	HARDNESS OF METAL minimum	TEST METHOD *
0.64 cm (0.25")	1.50 x 2.21 mm (0.059" x 0.087")	2.29 mm (0.090")	R30T63.0	ASTM E18
1.27 cm (0.50")	1.80 x 2.79 mm (0.071" x 0.110")	2.29 mm (0.090")	"	"
1.91 cm (0.75")	2.21 x 3.18 mm (0.087" x 0.125")	2.29 mm (0.090")	"	"
2.54 cm (1.0")	2.21 x 3.40 mm (0.087" x 0.134")	2.29 mm (0.090")	"	"
3.18 cm (1.25")	2.41 x 3.68 mm (0.095" x 0.145")	2.29 mm (0.090")	"	"
3.81 cm (1.5")	2.41 x 3.68 mm (0.095" x 0.145")	2.29 mm (0.090")	"	"
5.10 cm (2.0")	3.18 x 4.52 cm (0.125" x 0.178")	4.11 mm (0.162")	"	"
6.40 cm (2.5")	3.18 x 4.52 cm (0.125" x 0.178")	4.14 mm (0.163")	"	"
7.62 cm (3.0")	3.18 x 4.52 cm (0.125" x 0.178")	4.14 mm (0.163")	"	"

* Note for Table IV: When applicable, cylindrical corrections shall be in accordance with the applicable cylindrical correction table of ASTM E 18.

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TABLE V. COATED WOVEN COVER MATERIAL.

BINDER CAPACITY	COMPOSITION	WEIGHT OF FINISHED CLOTH minimum	BREAKING STRENGTH (GRAB METHOD) minimum	
			Warp Dir	Filling Dir
Greater than 1.3 cm (0.50")	1	288 g/m^2 (8.5 oz/yd^2)	200 Newtons (45 lb_f)	178 Newtons (40 lb_f)
1.3 cm (0.50") or less	2	172 g/m^2 (5.07 oz/yd^2)	156 Newtons (35 lb_f)	111 Newtons (25 lb_f)
Test Method		Fed.Std.191 Method 5041	Fed.Std.191 Method 5100	

TABLE VI. COVER BOARD

GRADES	BINDER CAPACITY	*THICKNESS	BASIS WEIGHT
I & II Flexible	Less than 2.5 cm (1")	0.89 mm (0.035")	0.59 kg/m^2 (120 lb_m/1000 ft^2) ± 5%
	2.5 cm (1") & Greater	1.27 mm (0.050")	0.83 kg/m^2 (170 lb_m/1000 ft^2) ± 5%
II & IV Stiff	3.8 cm (1-1/2") or Less	2.54 mm (0.100")	1.67 kg/m^2 (342 lb_m/1000 ft^2) -5%, +Unlimited
	Greater than 3.8 cm (1-1/2")	3.05 mm (0.120")	2.06 kg/m^2 (422 lb_m/1000 ft^2) -5%, +Unlimited
Test Method		TAPPI T 411	TAPPI T 410

* Thickness tolerance is +0.25mm (+0.010"), -0.13mm (-0.005").

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5. QUALITY ASSURANCE

5.1 Product conformance. The product provided shall meet the salient characteristics of this Commercial Item Description, conform to producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

Suppliers are encouraged to use quality control (QC) techniques that exhibit control over their processes (e.g. (Statistical Process Control) SPC techniques) as defined in American National Standards Institute (ANSI) Z1.1, Z1.2 and Z1.3/ American Society for Quality Control (ASQC) B.1, B.2 and B.3 that systematically reduce excess variations. Objective evidence shall be available that demonstrates overall measurement adequacy techniques and controls. These techniques shall ensure quality levels equal to, or greater than, those cited in applicable technical document or, herein.

End item inspection/testing may be used by the offerror or the Government representative, as a means to determine the effectiveness of the in process quality controls. In process controls shall not be substituted for end item performance testing.

For those characteristics for which there are no such controls, or for end item performance test, lot by lot sampling for inspection/test shall be required. Sampling shall be in accordance with the American National Standards Institute (ANSI)/ American Society for Quality Control (ASQC) Z1.4, Sampling Procedures and Tables for Inspection by Attributes

An inspection lot shall consist of all like items submitted for inspection at one time. The inspection level shall be as indicated in Table VII. The supplier must provide objective evidence (tests and inspection records) that the presented material meets the requirements of the sampling plan as indicated above.

5.2. Sampling. Samples shall be selected for inspection in accordance with the American National Standards Institute (ANSI)/ American Society for Quality Control (ASQC) Z1.4, Sampling Procedures and Tables for Inspection by Attributes, using the plans described in Table VII. A lot shall consist of a number of binders manufactured by the same process from the same components at the same time. Random samples shall be drawn from each lot for the end item inspection. Unless otherwise specified, every lot shall be tested

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TABLE VII.

Inspection	Inspection Level	AQL (%) defective	Sample Unit	Requirements Paragraph/s
Materials & Testing	S-1	4.0	1 Binder *	1A, 1B, 1C, 1D, 1E, 1F
Design & Construction	I	4.0	1 Binder	1G
Workmanship	I	4.0	1 Binder	1J

* Note for Table VII: For examination and testing of components and materials, the following components cannot be readily be tested in the end item form: coverboard, fiberboard, lining and covering materials, etc. Testing of these components must be in the pre assembled stage. These components must be traceable to identifiable lots of material, as defined by ANSI/ASQC Z1.4, and directly traceable to identifiable lots of finished product. Other components shall be end tested in a completed binder or, if necessary, disassembled items. In lieu of test data, certifications from suppliers may be accepted for raw material components, at the discretion of the Government.

6. PACKAGING, PACKING, AND MARKING: Packaging, packing, and marking shall be as specified in the contract or purchase order.

7. REFERENCED DOCUMENTS: The issue of the referenced documents in effect on the date of the solicitation for offers or request for proposals shall be used to determine conformance with the requirements of this Commercial Item Description.

7.1.1. CFR. Title 40, Code of Federal Regulations, Protection of the Environment, Part 247. Copies may be purchased with a VISA, Master Card or prepaid check through: Government Printing Office, Washington DC; Phone 202-783-3238. When ordering ask for: Title 40, Parts 190-250; Stock Number: 869-013-00-146-0.

7.1.2. Copies of Executive Order 12873, dated October 20, 1993 may be obtained from: The White House Publications Office, New Executive Office Building, White House,Att: Publications, Washington, DC. 20500 (202) 512-1800.

7.1.3. (ANSI)/ American Society for Quality Control (ASQC) Z1.4, Sampling Procedures and Tables for Inspection by Attributes.

Application for copies can be obtained from American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.

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7.1.4. Federal Standards And Test Methods:

Federal Standards, Specifications and Commercial Item Descriptions, may be obtained from the General Services Administration Specifications Section (3FBO-W) in Suite 8100 at 490 L'Enfant Plaza, SW Washington D.C.20407. phone numbers(202) 708-9205, or FTS-458-9205.

7.1.5. American Society for Testing and materials (ASTM).

Available from: American Society for Testing and Materials; 1916 Race Street; Philadelphia, PA 19103.

7.1.6. Technical Association of the Pulp and Paper Industry (TAPPI):

TAPPI Test Methods.
Available from Technical Association of the Pulp and Paper Industry; Technology Park/ Atlanta; P.O. Box 105113; Atlanta, GA 30348.

7.1.7. National Association of State Textbook Administrators (NASTA):

NASTA Type II - Reinforced Paper: Polymer or Resin Reinforced Cellulose Fiber Webs.
NASTA Type III - Synthetic Fiber Structure: Strands of Interconnected Fibers.

Application for copies for specification for NASTA Types II and III should be addressed to the Advisory Commission on Textbook Specifications; 45 William Street, Wellesley, MA 02181-4004.

8. ORDERING DATA. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

8.1. Grade, sheet dimension, capacity, number of rings, and ring spacing.
8.2. Color of Binder.
8.3. End or Side opening.
8.4. Booster or other device (1) opens the rings; or (2) opens and closes the rings; or (3) opens, closes, and locks the rings.
8.5. Whether special features (e.g. pockets, metal hinges, labelholders, labels, etc.) are required.
8.6. Whether special marking, lettering or labeling of binder is required.
8.7. Packaging and packing required.

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MILITARY INTEREST:

NONE: DoD has no registered interest in revisions and amendment to this Commercial Item Description until further notice.

PREPARING ACTIVITY.

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